

Feasibility study and optimization of a standalone hybrid system for remote area: A case study of High Atlas mountains in Morocco.

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Abstract—Hybrid systems based on photovoltaic and wind energy sources are more recognized as cost-effective generation sources for remote rural area and isolated power system. This paper presents a detailed feasibility study and techno-economic evaluation of a standalone hybrid system, based on renewable energy (RE) sources with batteries for a remote area installed in High Atlas mountains in Morocco. The HOMER (Hybrid Optimization of Multiple Energy Resources) software was used to simulate the system and perform a techno-economic evaluation. A description and analysis of the proposed system are presented, net present cost (NPC), cost of energy (COE) are evaluated to achieve an optimal configuration. A sensitivity analysis on load consumption was performed to evaluate the impact on COE, NPC, and other parameters. By adopting the solution proposed in this paper, the load will be supplied in all conditions. Additionally, this solution will not only provide electricity to people of Atlas who already suffer from isolation but also to raise their standard of living.

Index Terms—Hybrid systems; Remote area; Solar-wind-battery system; techno-economic evaluation; HOMER; Net present cost (NPC); Cost of energy (COE).

I. INTRODUCTION

For many centuries, energy has been considered as an essential basis for all social activities, it becomes a part of human lives. On the other hand, energy is still to be the biggest crisis to human beings, Nowadays, most of the energy used on earth comes from fossil fuels. According to the recent exploring and exhausting rate, excessive consumption of these resources could lead to depletion in a few decades. In addition, these factors cause global warming too. In these territories, there are still about 1.5 billion inhabitants in the world still having no access to electricity [1], [2].

Over many decades, scientists and engineers around the world have been supporting the utilization of locally available renewable energy resources to develop remote rural areas, especially in developing countries. In Morocco, the Atlas region is one of highest places, it is nearly 1900 meters above sea level, furthermore, Atlas contains the most rural areas. The energy supply for those villages is commonly powered by diesel generators (DG). However, people there feel more and

more depressed especially by the unacceptable rise in diesel prices and extra costs of shipment. Moreover, establishing a grid extension in high and isolated areas is very difficult and much expensive.

The rising cost of diesel fuel and the rapidly declining cost of renewable energy technologies make energy supply by renewables a real competitive with conventional energy [3]. Therefore, by using a combination of renewable energies, electrification of remote and isolated sites remains a problem of the past.

Several publications have appeared in recent years documenting the field of system modeling, simulation, component sizing, economic analysis, and particularly system optimization [4], [7], [8]. To carry out such research, a simulation needs to be performed. Several software are available for the study of the feasibility and the optimization of the hybrid system [9], [10]. Among the simulation tools, HOMER software is one of the most widely used for standalone renewable energy systems (RESs) [4]. This software helps to design RESs and facilitate the comparison of power generation technologies across a wide range of applications.

This paper focuses on studying the feasibility of using a combination of solar and wind energy to meet the electricity requirements for this remote area in conjunction with the battery storage.

The solar radiation, wind speed and ambient temperature data on this area are provided by NASA surface meteorology and solar energy [1], then renewable energy resource and the benefit of hybrid solar and wind system have been evaluated. HOMER software is employed as a platform for sizing and optimizing the proposed system [6]. This paper is organized as follows. Section II describes the configuration and control of the proposed hybrid system. The modeling of the system is detailed in Section III within the modeling of system component: PV, wind, and batteries while synthesizing load profile. The last section presents the simulation results, thousands of cases in hourly basis are carried out and compared with regards to system net present cost (NPC) and cost of energy (COE).

Moreover, a sensitivity analysis on load and renewable energy resource has been done.

II. SYSTEM DESCRIPTION

Hybrid renewable energy systems (HRES) based on PV-wind power generation systems can efficiently improve energy usage on the system, supply effectively the energy, while a good optimization of the system may be sufficient to reduce the storage element which is considered the most expensive. According to several publications in the past few years, HRES based PV-wind systems is techno-economically the most adapted and viable for rural electrification [11], [12].

A. System configuration

Fig. 1 presents system architecture and energy flow for the proposed hybrid solar-wind system with batteries [13]. The system essentially consists of PV array, wind turbine (WT), battery bank, inverter, controllers, and other devices. By utilizing a common DC bus, the generated power from primary sources is converted into AC by the inverter. The energy exchanged on the system must be equilibrated, after supplying the loads the available excess energy is directly fed into the batteries. In the case when the batteries are fully charged and there is no more energy demand, the surplus energy is dumped. In the second case, the batteries release power to the load when the generated output power is unavailable or is insufficient to supply the load.

The combination of several sources with storage system is highly recommended for a better exploitation of energy and supply load in all weather conditions and load fluctuation.

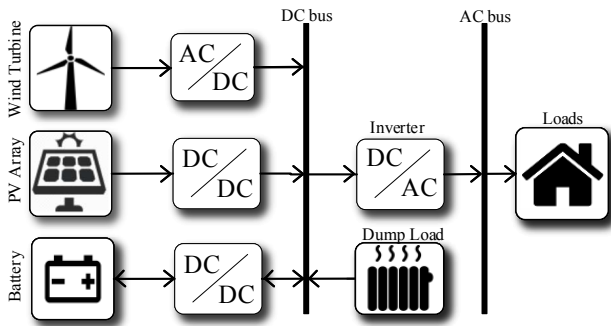


Fig. 1. . The energy flow diagram of the proposed hybrid solar-wind system.

B. System control

Decent control of hybrid energy systems with multiple RE/conventional-DGs and energy storage is crucial to improving system reliability and operation performance. The studied system can be controlled using different methods. In this case, the only dispatchable source is the battery bank. When the net load, which is the difference between the actual load and the renewable energy output, is negative, the surplus energy is used to charge the batteries. In the second case, when the net load is positive, the batteries release energy to satisfy the load. The flowchart of Fig . 2 explains the control strategy established for this system.

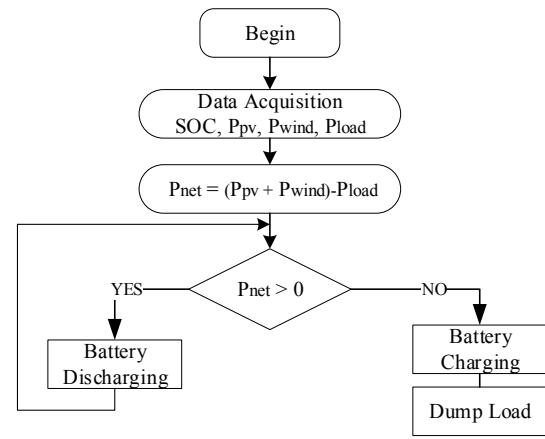


Fig. 2. Control strategy of this hybrid solar-wind system.

III. SYSTEM MODELING

A. Remote area load profile

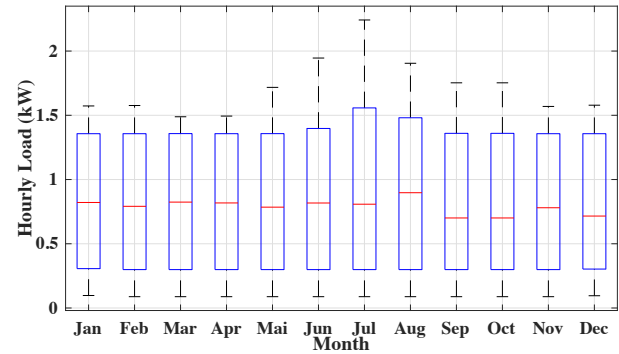


Fig. 3. Hourly load during a year in this area.

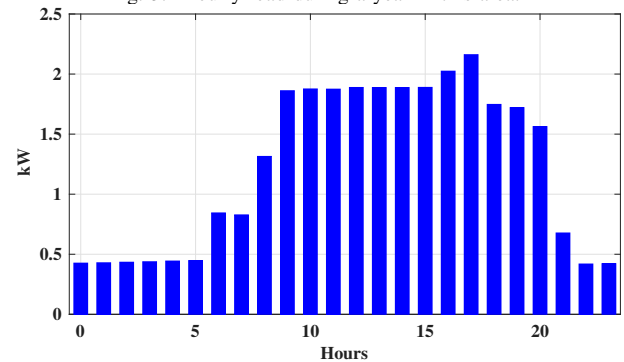


Fig. 4. Hourly load during a day in this area.

In this work, a feasibility study of the HRES based PV-wind systems in a remote area of Atlas regions in Morocco has been developed. The daily base load in this area was estimated at 20 kWh/day. In order to correctly model the daily and annual consumption of the site, the average consumption was taken into consideration then some randomness was added for different days and months, to create a considerably moderate profile for this site within a year. The hourly load profile during

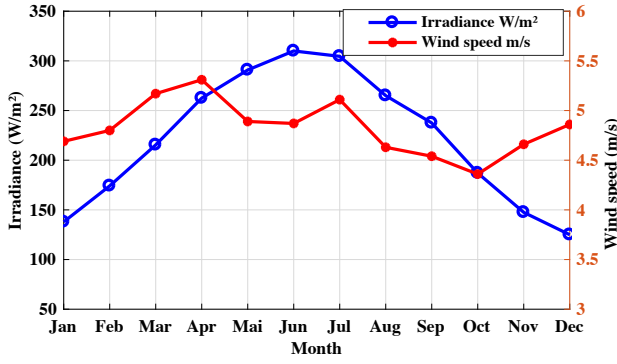


Fig. 5. Monthly solar and wind energy resource on Atlas.

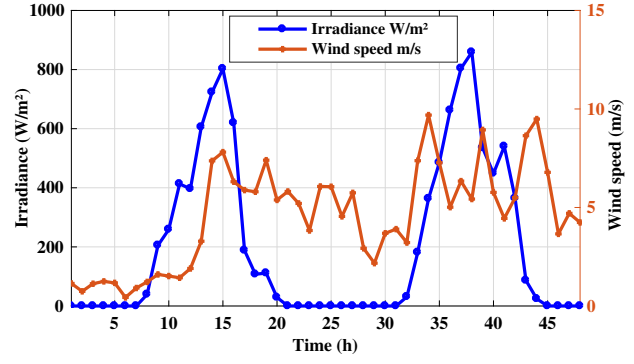


Fig. 6. Daily solar radiation and wind speed distribution on Atlas.

different months is illustrated in Fig. 3. From this figure, it can be noticed that load consumption is highly important in the winter, taking into consideration that the Atlas is considered one of the coldest regions in Morocco, these results can be contributed by the high usage of the cooling machine in the winter. The daily load profile is represented in Fig. 4 by a series of powers, each considered as constant over a time step of 1 hour during 24 hours.

B. Renewable energy resource

Table I shows the different geographical coordinates of the selected site.

TABLE I
GEOGRAPHICAL COORDINATES OF THE SELECTED SITE

Site	Latitude	Longitude	Altitude (m)
Anfgou (High Atlas)	32.287.957	-5.392.646	1935

The weather data on this site was provided by NASA surface meteorology and solar energy [5], including solar radiation, wind speed, and ambient temperature. The mean solar and wind energy resources by month are presented in Fig. 5, while, Fig. 6 shows the hourly wind and irradiation profile during two different days in April 2005. Those figures explain the good complementary nature of PV and wind systems. The night provides a good wind resource but poor solar irradiation, while the opposite reasoning can be considered for the day. Based on these results, it can be concluded that coupling multiple RES can provide better performance and great utilization of energy.

C. System components information

1) *PV modules*: The cost of solar panels has decreased significantly over the last decade and are currently selling for 1.5€/W. Fig. 9 summarizes basic information about the chosen PV module. The total power of the photovoltaic module is considered 1 kW and the efficiency of the module in stand test condition (STC) is 14.3%. The initial capital cost of a PV module was assumed as 1.8€/Wp including the PV module and array installation cost. The replacement cost was estimated to be the same as the initial cost since the photovoltaic panel

can be viewed as a single solid element. The operating and maintenance (O&M) cost was assumed to be zero since it is negligibly small.

2) *Wind turbine*: Unlike the photovoltaic technologies, the wind turbine is still expensive. The initial cost of the selected wind turbine as presented in Fig. 9 was estimated at 3.0€/Wp, the replacement cost was considered to be the same as the initial cost. The operating and maintenance (O&M) cost was assumed to be 3.0€/Wp. The power curve of the wind turbine is presented in Fig. 7.

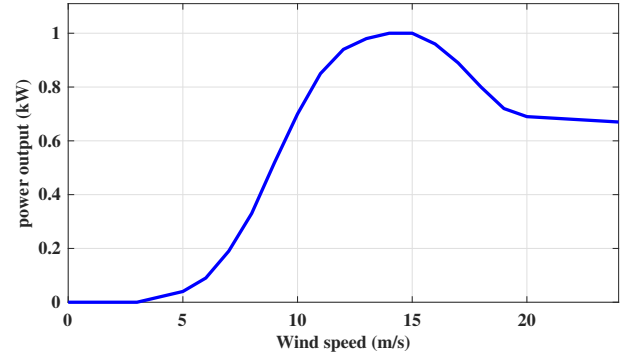


Fig. 7. Power curve of the wind turbine.

3) *Battery bank*: The sizing approach used to determine the required number of batteries is shown in Eqs. (1)-(3).

$$C_{Ah} = \frac{E_t}{\eta_B \cdot DOD \cdot V_B} = \frac{N_{day} \cdot E_{Load}}{\eta_B \cdot DOD \cdot V_B} \quad (1)$$

$$N_{battery} = \frac{C_{Ah}}{C_{single}} \quad (2)$$

$$N_{string} = \frac{\eta_{battery}}{48/V_B} \quad (3)$$

where E_{Load} is the daily energy consumption (20 kWh), E_t is the total of energy demand for the continuous number of autonomous days, N_{day} is the number of autonomous days powered only by the batteries, $N_{battery}$ is the total number of batteries, N_{string} is the string number, η_b represents the battery and inverter efficiency, V_B is the battery voltage, DOD

is depth of discharge and C_{single} is the storage capacity of a single battery. The sum of the voltages provided by each string should equal the nominal system voltage of the DC bus (48 V).

In order to protect the batteries against the deep charge and discharge, a parameter called state of charge (SOC) was introduced in this study. When the SOC is between SOC_{min} and SOC_{max} , the RES will continue charging the batteries while they start discharging as the load can't be anymore supplied by primary sources. The surplus energy is dumped when the batteries are saturated.

IV. RESULTS AND DISCUSSION

A. System physical configuration

Simulation of the hybrid system has been conducted using HOMER. Based on the meteorological data and load consumption, many cases were carried out to reach an optimal system configuration. The simulation results displayed in Fig. 9 show that the optimal system contains PV array (10,2 kW), WT (1 units, 1 kW), batteries (44 units of 1 kWh, a total of 44 kWh) and converter (4.74 kW).

B. Operating performance of system components

Table II, presents operating performance and economic results of the studied system. The capacity factors of PV array and WT were relatively low because of massive amounts of the wasted energy. The levelized costs for the PV and WT were respectively 0.0864€/kWh and 0.453€/kWh. Knowing that the excess energy in the system is used in charging the batteries, the energy cost of the batteries was zero. The battery wear cost of 0.419€/kWh was significant, nearly 72% of the COE, while the battery cost (28 968€) was even higher than that of the main renewable source which is the PV module (19 670€). That results can be explained by the high cost of any storage system. In certain cases, some users may override or discard the use of batteries and introduce more renewable energy generators. Nevertheless, this solution may result in a large loss of energy.

Fig. 10 presents the monthly average electric production from PV array and WTs. The PV provides almost 95% of the total production during the simulation year while the WTs provide only 5%. The PV output was extremely high in the summer months, especially in July. The wind energy was less important in summer and during days but found to be significant in April, March and especially at nights.

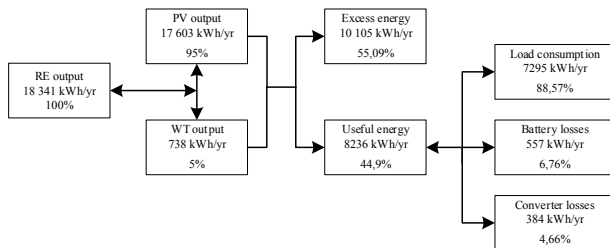


Fig. 8. Summary of the energy flow during the simulated year.

C. Total cost break-down

The estimated cost by components and cost type is shown in Fig. 11. The initial cost (IC) and total net present cost (NPC) is respectively 35 975€ and 54 873,64€. The corresponding levelized cost of energy (COE) was 0.585€/kWh, approximately seven-time less than diesel generator based mini-grid. During the simulation, the updated price of the fuel in Morocco is taken into account which is estimated at 0.88€/L at this moment. Table III represents the difference in COE and NPC between several possible configurations.

Combining multiple sources of energy is considered a prominent solution regarding COE. In the past, in remote areas, diesel generators and grid extension by cables was the only solution for the electrification. In addition, by using DGs, the COE becomes considerably higher and it costs 3,98€/kWh. However, considering the proposed solution, the COE drops to 0,582€/kWh approximately seven-time smaller. It should be noted that the cost of the batteries is nearly 52% of the total NPC. This indicates that the cost of a storage subsystem in a stand-alone hybrid renewable system to be dominant.

D. Energy flow analysis

A summary of the energy flow of the system is presented in Fig. 8. As it can be noticed, 95% of the renewable energy output was generated by the PV array and 5% by WTs. The excess energy is so important, about 55% of total produced energy, this portion is then transferred to the dump load. These results can be interpreted by the low demand of energy or the lack in terms of storage capacity, for example, in summer weekdays, the generated power is significantly high but conversely, the load consumption in the middle of the day is low. Among the total useful electricity (8 236 kWh), 88,5% was consumed by the users, 4.9% for battery losses and 9.5% for converter losses.

E. Hourly simulation results

Fig. 12 represents the hourly simulation results during four consecutive days of April. The wind energy resource was extreme at night and during the early morning hours although the solar energy was available during the day. The green line illustrates the load profile for this site. The excess electricity was managed to charge the batteries. The red line illustrates the battery state of charge (SOC), it is managed according to the flowchart explained in Fig. 2. In order to demonstrate the effect of load demand and weather condition on SOC, specific cases are taken into consideration. During 1st and 2nd April, the sources are able of supplying the load and the batteries were periodically charged or discharged. The SOC values dropped during the third and fourth day due to poor wind speed. It is found that the lowest value of 68% occurred on the fourth day, but still much higher than the allowable minimum SOC.

F. Sensitivity analysis on load consumption

The daily load consumption in the base case was assumed as 20 kWh. To simulate the effect of the load consumption

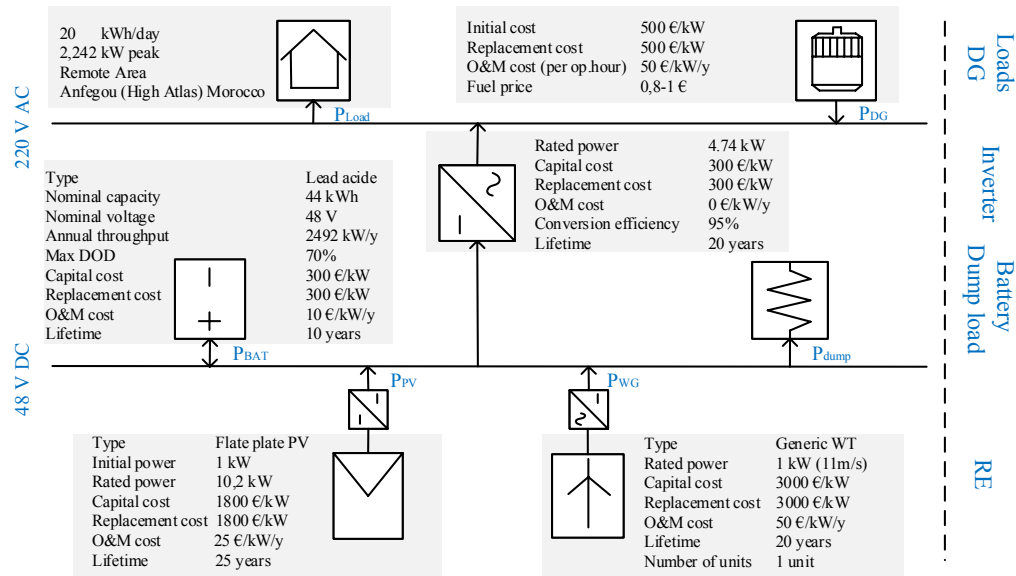


Fig. 9. Physical system configuration of the proposed hybrid solar and wind system with batteries in HOMER.

TABLE II
OPERATING PERFORMANCE AND ECONOMIC RESULTS OF THE STUDIED SYSTEM

Battery			PV		WT		
Parameters	Value	Units	Parameters	Value	Units	Value	Units
Batteries number	44		Rated Capacity	10,2	kW	1	kW
String Size	4		Mean Output	2,01	kW	0,0842	kW
Strings in Parallel	11		Capacity Factor	19,7	%	8,42	%
Storage Wear Cost	0,419	€/kWh	Total Production	17603	kWh/yr	738	kWh/yr
Expected Life	10	yr	Hours of Operation	4377	hrs/yr	6035	hrs/yr
Energy In	2782	kWh/yr	Levelized Cost	0,0864	€/kWh	0,453	€/kWh
Energy Out	2229	kWh/yr	Total cost	19969,7	€	4322,34	€
Energy Cost	0	€/kWh					
Total cost	28968,4	€					

TABLE III
DIFFERENT CONFIGURATION POSSIBLE FOR 20 kWh OF THE SITE

Configuration Type	PV size (kW)	WT number	Battery Strings	DG size (kW)	Total NPC (€)	COE (€/kWh)
PV/Wind/Battery	10,2	1	44		54 874	0,582
PV/Battery	11,7		48		55 197	0,585
PV/Battery/DG	6,05		20	15	101 578	1,08
PV/Wind/Battery/DG	6,2	1	20	15	103 128	1,09
DG				15	375 140	3,98

on COE, NPC and energy excess, a simulation of different load consumption varying from 5 to 120 kWh/day has been performed.

Fig. 13 presents the optimal system configurations, COE and NPC under different load levels. The COE and NPC of PV-Wind-Battery, PV-Wind and PV-DG-Battery system respectively are in the range of 0.47€/kWh-0.58€/kWh and 13 824€-267 968€. From this figure, it can be seen that the increase in the load generates a progressive decrease of COE and energy excess, this supports the results obtained in the previous sections. As may be seen in the case of 80 kWh, by using a combination of DG, PV and Battery, the excess of

energy decreases considerably from 56% to 21%, this proved by the introduction of DGs which provide energy on demand. In addition, the economic benefits of PV-DG-Battery system becomes more significant the greater is load consumption.

The increase of wasted energy between 50 kWh and 60 kWh is justified by the expansion of the wind turbine numbers (1 WT when load consumption is less than 50 kWh and 2 WTs on the other case).

The COE is significantly higher in low load consumption, this result may be justified by the initial cost of the components, more the quantity of components increases, more the price-quality ratio decreases.

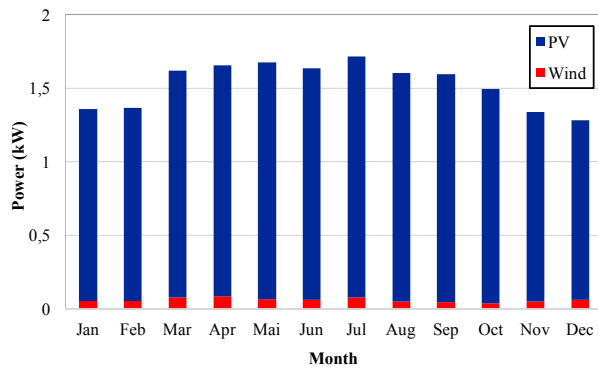


Fig. 10. Daily mean power generated by PV and wind in different months.

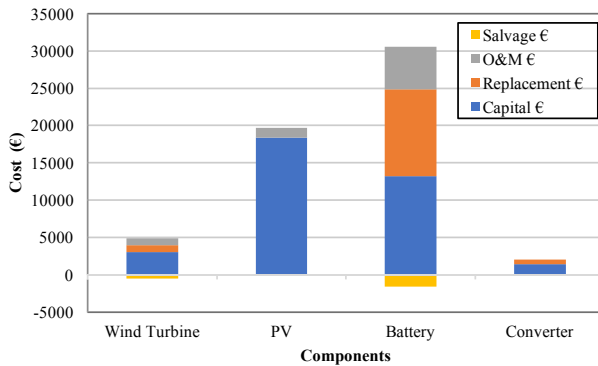


Fig. 11. Cash flow break-down by components and cost type.

V. CONCLUSION

This study shows the utility of renewable energies in the electrification of the remote area and very precisely in Morocco. From the results obtained, it is easy to justify the preference of electrical power generation systems based PV-wind with batteries instead of the diesel generators. Investing in such technology can be fruitful in the long term. The combination of solar-wind energy and batteries can supply continuous power to this remote area while benefiting from a fairly low energy cost. The optimal system essentially comprises 10,2 kW PV array, 1 kW WT, 44 batteries and a 4.74 kW converter. The timing mismatch between production and consumption of energy lead to a considerably huge amount of waste of the energy. Despite these losses, the proposed solution improves two important parameters, NPC and COE which respectively correspond to 54 873,64€ and 0.582 €/kWh. However, generation systems based DG are still widely used but with expected continuing rapid development in the renewable energy industry and upgrades in storage technology, the off-grid system's based RE cost will be reduced and remote areas based on renewable energy grid could be more promising.

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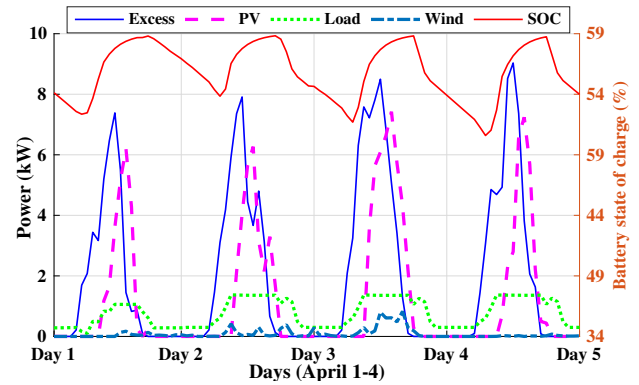


Fig. 12. Hourly simulation results for four consecutive days (1st to 4th April).

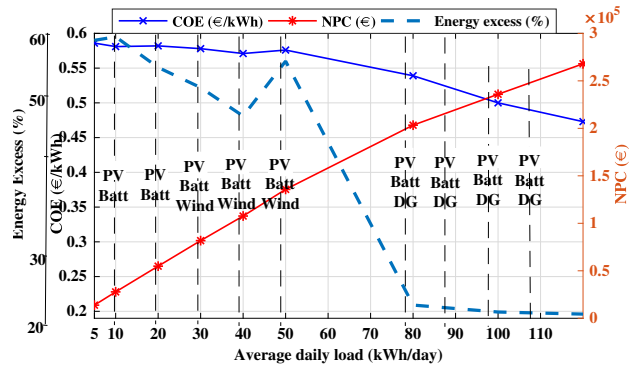


Fig. 13. COE, NPC and energy excess for different system types against load demand.

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